

$$\frac{4. a}{X = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{bmatrix}_{n \times d} \quad \vec{y} = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}_{n \times 1}$$

$$J_{\lambda}(B) = \frac{1}{2} \|XB - \vec{y}\|_2^2 + \frac{\lambda}{2} \|B\|^2$$

a) prove that $B_{\lambda} = (X^T X + \lambda I_{d \times d})^{-1} X^T \vec{y}$ (minimizer)

$$J_{\lambda}(B) = \frac{1}{2} \|XB - \vec{y}\|_2^2 + \frac{\lambda}{2} \|B\|^2$$

Can be written as

$$J_{\lambda}(B) = \frac{1}{2} (XB - \vec{y})^T (XB - \vec{y}) + \frac{\lambda}{2} B^T B$$

$$= \frac{1}{2} [(B^T X^T - \vec{y}^T) (XB - \vec{y})] + \frac{\lambda}{2} B^T B$$

$$= \frac{1}{2} [B^T X^T X B - \underbrace{B^T X^T \vec{y} - \vec{y}^T X B + \vec{y}^T \vec{y}}_{\text{scalars & transposes are equal}}] + \frac{\lambda}{2} B^T B$$

ie $B^T X^T \vec{y} = \vec{y}^T X B$

$$J_{\lambda}(B) = \frac{1}{2} (B^T X^T X B - 2B^T X^T \vec{y} + \vec{y}^T \vec{y}) + \frac{\lambda}{2} B^T B$$

$$J_{\lambda}(B) = \frac{1}{2} B^T X^T X B - B^T X^T \vec{y} + \frac{1}{2} \vec{y}^T \vec{y} + \frac{\lambda}{2} B^T B$$

Let's compute gradient of $J_{\lambda}(B)$ w.r.t B .

$$\nabla J_{\lambda}(B) = \frac{1}{2} (2X^T X B) - X^T \vec{y} + 0 + \frac{\lambda}{2} \cdot 2B \cdot I$$

$$\nabla J_{\lambda}(B) = X^T X B - X^T \vec{y} + \lambda I_{d \times d} B$$

4.a

$$\nabla J_{\lambda}(\beta) = X^T X \beta - X^T \bar{y} + \lambda I_{d \times d} \beta$$

For closed form, we set $\nabla_{\lambda} J(\beta) = 0$ to find the minimum value (minimizer), therefore.

$$X^T X \beta - X^T \bar{y} + \lambda I_{d \times d} \beta = 0.$$

$$(X^T X + \lambda I) \beta = X^T \bar{y}$$

$$\boxed{\beta = (X^T X + \lambda I_{d \times d})^{-1} X^T \bar{y}}$$